Dark Web Anonymity and Cryptocurrency

EC Council Course Completion Report

Subject Code: CSC 2735

Subject Name: Network and Data Security

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Course: Introduction to Dark Web, Anonymity, and Cryptocurrency

Provider: CodeRed by EC-Council

Certificate Number: 521166

Course Duration: 1 hour

Certificate Issued: 19 August 2026

Report Date: 3 November 2026

This report explains the purpose of the dark web, the main concepts associated with anonymity and cryptocurrency, and their relevance to network and data security. The attached course completion certificate records completion of the introductory course.

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1.0 Introduction and Objectives

The dark web is a part of the internet that uses specialised networks and software to make services available while protecting aspects of user and server identity. It is relevant to cybersecurity because the same privacy mechanisms can support legitimate confidential communication and be misused for fraud, stolen information and other harmful activity.

The course named on my certificate is Introduction to Dark Web, Anonymity, and Cryptocurrency. This report connects these three topics to CSC 2735 Network and Data Security. Its main lesson is that privacy depends on technology, user behaviour and the security of the device, rather than on one browser or payment method alone. [1-4]

1.1 Objectives

• Distinguish the surface web, deep web and dark web.

• Explain why onion services support private communication and access to information.

• Describe the role of Tor Browser and the limits of anonymity.

• Explain the relationship between cryptocurrency, pseudonymity and public transaction records.

• Identify common security risks and suitable precautions.

• Document course completion accurately using the supplied certificate.

2.0 Core Concepts

2.1 Surface web deep web and dark web

The surface web contains public pages that ordinary search engines can index. The deep web includes material outside normal search indexes, such as private email, university portals and account pages. The dark web is a smaller category of services available through specialised networks. A password-protected student portal is therefore part of the deep web without being a dark web service.

2.2 The purpose of the dark web

There is no single purpose shared by all dark web services. A major legitimate use is to publish and exchange information while reducing exposure of identifying network details. Onion services can support confidential communication, anonymous publishing and access to information in restrictive environments. These benefits also create opportunities for abuse, so an anonymous service should still be assessed for authenticity and risk. [2,3]

2.3 Tor Browser and onion services

Tor routes connections through multiple relays to reduce the direct link between a user and a destination. Tor Browser is configured to use that network and includes privacy protections. Ordinary websites can be visited through Tor; onion services are a separate service type reached inside the Tor network. Their addresses end in .onion. For onion services, the connection stays within Tor instead of using a normal exit relay. [2,3]

Tor reduces some forms of tracking but does not guarantee complete anonymity. Signing into a personal account identifies the user to that service. Malware, identifying content and unsafe handling of downloaded documents can also undermine privacy. Encryption and anonymity address different questions: encryption protects content, while anonymity seeks to reduce links to an identity.

2.4 Cryptocurrency and Bitcoin

Cryptocurrency uses cryptographic keys to authorise transfers recorded by a network. A wallet manages keys; it does not make its owner automatically anonymous. Bitcoin transactions are public, and addresses may be linked to people through reuse, published information or other records. Bitcoin is better described as pseudonymous than as completely anonymous. Private keys and recovery phrases must remain secret. [4]

3.0 Learning Outcomes and Reflection

The following learning summary explains the central ideas of the course topic and their application to network security.

3.1 Understanding the dark web

The most useful distinction is between hidden content and private network access. Content may be absent from a search engine simply because it requires a login. Onion services use a different access mechanism. Recognising this difference prevents the assumption that everything outside ordinary search results belongs to the dark web.

3.2 Understanding the limits of anonymity

A key lesson is that hiding a network address does not remove every identifying clue. A real name, personal account or reused username can connect activity to an individual. This links the topic to operational security: privacy depends on consistent choices as well as software configuration. Tor Browser also protects its own traffic; its presence does not mean every application on the device is using Tor. [2]

3.3 Understanding cryptocurrency privacy

Public blockchain records can preserve transaction history even when an address has no name displayed beside it. This changes how cryptocurrency should be discussed in security investigations. An address alone does not prove the identity of a person, but it may become relevant when combined with reliable supporting evidence. [4]

3.4 Relevance to Network and Data Security

These topics connect to confidentiality, traffic analysis, endpoint security, authentication and evidence handling. An encrypted route does not make a malicious destination trustworthy. Similarly, a confidential connection cannot protect data after the user sends it to a fraudulent website. Security analysis must consider the device, network, service and human decisions together.

3.5 Personal reflection

My course completion is a starting point for further study of online privacy and security. The main takeaway is to evaluate claims of anonymity carefully and to separate technical protection from trust in a website. The later demonstration documented in Section 6 connects these concepts to visible browser behaviour. Further independent practice would strengthen my understanding of the underlying network mechanisms.

4.0 Knowledge Application Exercise

This is a worked, desk-based exercise included to apply the report concepts. It is not a record of a completed browser session, an onion-site visit or a cryptocurrency transaction.

4.1 Classifying internet services

Scenario: A public university news page, a student email account and a Tor onion service need to be classified.
Analysis: The public news page is surface-web content if indexed. The email account is deep-web content because access is restricted. The onion service is a dark-web service because it requires the appropriate network access.
Learning demonstrated: Search visibility and access technology are different classification criteria.

4.2 Evaluating an anonymity claim

Scenario: A person uses Tor Browser and then signs into a personal account.
Analysis: The service can associate the session with the account. Tor may reduce exposure of the originating network address to the service, but account authentication still establishes an identity relationship.
Learning demonstrated: Network privacy does not remove information voluntarily disclosed to a service.

4.3 Evaluating a cryptocurrency claim

Scenario: Someone says a Bitcoin payment is invisible because the address has no real name.
Analysis: The payment is still recorded publicly. A lack of an attached name does not prevent later linkage through other evidence.
Learning demonstrated: Pseudonymity differs from invisibility and guaranteed anonymity. [4]

4.4 Practical demonstration on the student computer

On 3 November 2026, an assistant-operated demonstration was carried out on the student computer with permission. The existing Tor Browser was opened, a connection to Tor was established, the official connection-check page was visited, the relay circuit was inspected, and the Tor Project community onion service was loaded. Screenshots and results appear in Section 6. This later demonstration is separate from the course completion recorded on 19 August 2026.

5.0 Security Analysis and Recommendations

5.1 Phishing and impersonation

A convincing page can still be fraudulent, and an onion address does not establish trust by itself. Use addresses published by the organisation through its verified official channels. Avoid entering credentials or sharing personal details on an unfamiliar service.

5.2 Malware and endpoint exposure

Privacy tools cannot compensate for a compromised device. Keep the operating system and browser updated, avoid unknown executables and browser extensions, and treat downloaded documents cautiously. A document opened in another application may use a different network route and expose information outside the browser session.

5.3 Identity linkage and information disclosure

Personal accounts, reused identifiers and revealing uploads can connect activity to a person. Consider what information is disclosed before using a service. Do not assume that a proxy, Tor connection or private browsing window removes all identifying signals.

5.4 Cryptocurrency loss and disclosure

A public blockchain should be treated as a visible transaction record. Protect wallet recovery material, verify recipient information and understand the consequences before authorising a transfer. No wallet creation, purchase or transfer is needed to demonstrate the introductory concepts in this report. [4]

5.5 Ethical practice and evidence quality

Learning should use legitimate resources and authorised environments. Accessing private systems, collecting other people's data or testing a service without permission is outside the scope of this report. Evidence should record what actually occurred, preserve relevant context and distinguish observations from interpretations. A course certificate establishes a completion record; it does not by itself demonstrate independent practical proficiency.

6.0 Certificate and Practical Evidence

Figure 1 reproduces the supplied CodeRed completion certificate. It names Mohamed Hussein, identifies the course as Introduction to Dark Web, Anonymity, and Cryptocurrency, and records certificate number 521166, a duration of 1 hour, and an issue date of 19 August 2026.

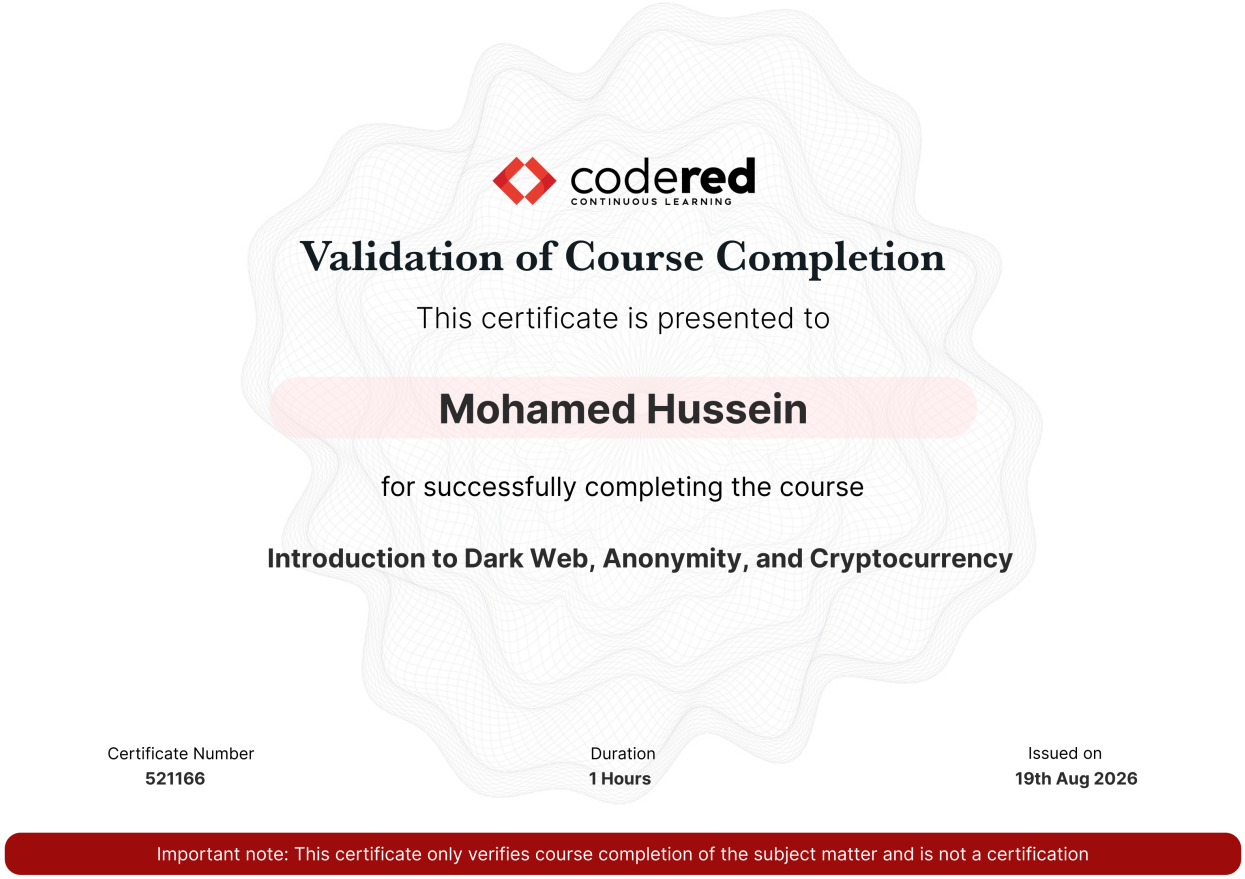


Figure 1 CodeRed validation of course completion supplied by the student.

The certificate states that it verifies course completion and is not a professional certification. It provides documentary evidence of completion, rather than evidence of a particular practical activity. The exercise in Section 4 demonstrates the application of concepts within this report and is separate from the certificate.

6.1 Practical Setup and Connection

Activity date: 3 November 2026. These screenshots were captured directly from an assistant-operated session on the student computer, with permission. They document a new practical demonstration and are separate from the August course completion record.

Environment: Existing Windows desktop installation of Tor Browser. The executable metadata reported version 140.17.0. No additional software installation was required. The session used public Tor Project resources and did not create accounts or carry out cryptocurrency transactions.

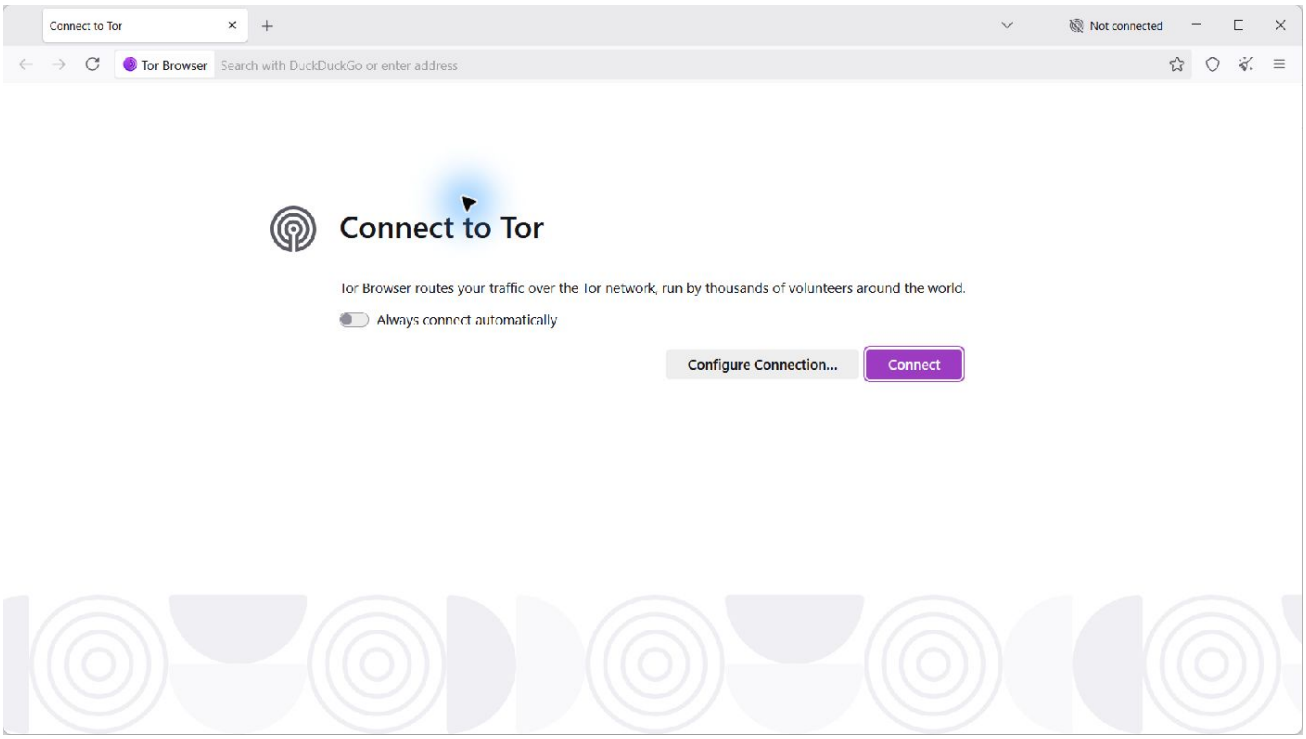


Figure 2 Tor Browser before connecting. The page displays Connect to Tor and a Not connected status.

Procedure: Open the existing Tor Browser shortcut, select Connect and allow the connection process to finish. Then visit https://check.torproject.org/ to test whether this browser session is using Tor.

Observation: The browser completed the connection process and the check page returned a successful result, shown in Figure 3. Merely opening the application would not have established this result.

6.2 Connection Verification and Relay Circuit

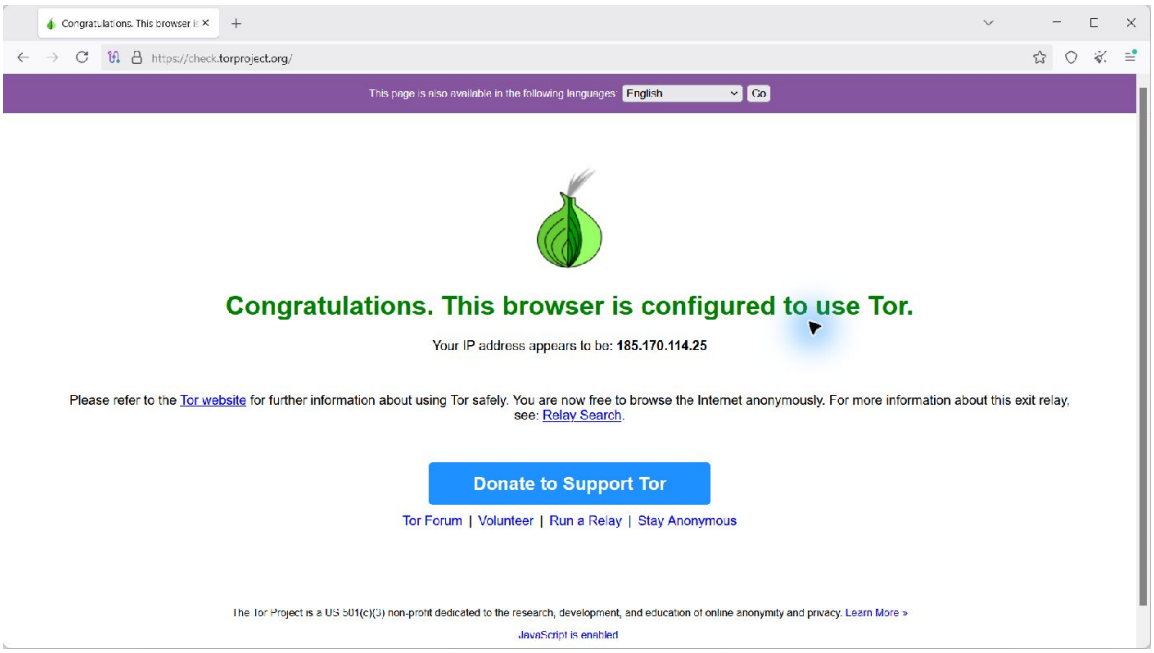


Figure 3 The official Tor connection check reports that the browser is configured to use Tor. The address displayed by the page is the observed exit relay address, not evidence of the original device address.

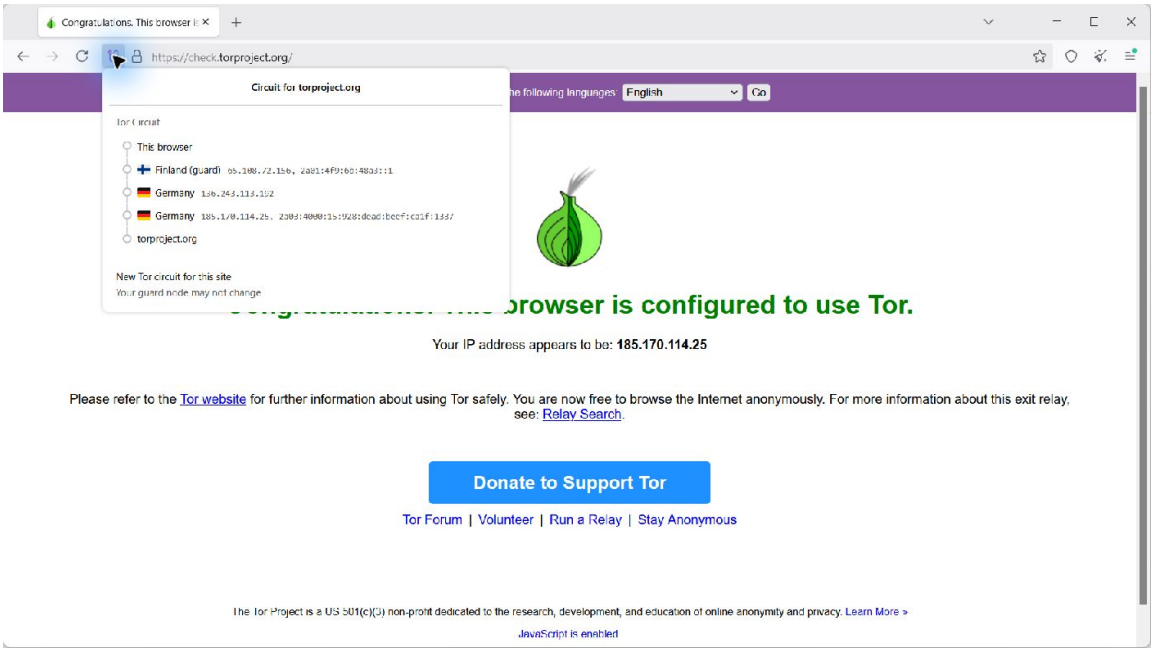


Figure 4 The browser circuit panel shows a guard in Finland followed by two relays in Germany for this website connection.

Interpretation: The check confirms that this request used a recognised Tor exit. The circuit panel illustrates the multi-relay route. These observations do not establish complete anonymity or prove that other applications on the computer use Tor.

6.3 Access to an Official Onion Service

The onion address for community.torproject.org was checked against the Tor Project official service directory at https://onion.torproject.org/. The corresponding onion service was then opened in Tor Browser. [6]

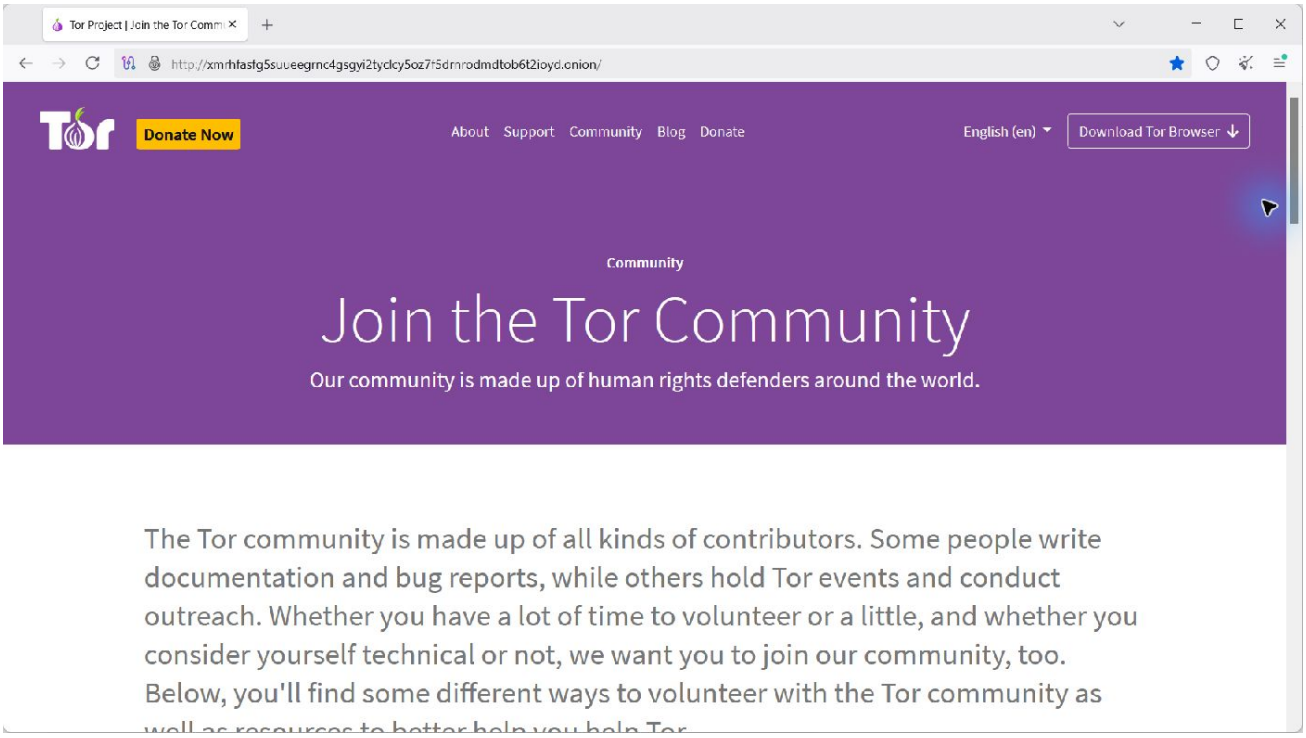


Figure 5 The Tor Project community page loaded successfully at its .onion address in Tor Browser. The address bar remains visible to identify the type of service accessed.

Result: This demonstrates access to a legitimate onion service. Unlike the ordinary HTTPS connection in Figure 3, onion-service traffic is carried within Tor without using a normal exit relay. The example shows a lawful informational use of dark web technology.

Evidence limits: The screenshots establish the visible browser states captured during this demonstration. They do not prove independent proficiency, previous course lab work, access to other onion services or completion of any cryptocurrency activity. No claim of a security vulnerability or penetration test is made.

Learning connection: Figures 2 to 5 show the sequence from an unconnected browser to a verified Tor connection and then an onion service. They provide practical context for the concepts in Sections 2 and 3.

7.0 Conclusion

The dark web is relevant to network and data security because it illustrates how privacy mechanisms can protect communication while also complicating trust and accountability. Tor and onion services help reduce exposure of network identities, but user behaviour and endpoint security remain important. Cryptocurrency introduces another distinction: an address can be pseudonymous while its transactions remain public.

Completion of Introduction to Dark Web, Anonymity, and Cryptocurrency is documented by the attached certificate. The learning summary and worked exercise connect the topic to confidentiality, identity exposure and evidence quality. The later assistant-operated practical demonstration records a successful Tor connection, inspection of a relay circuit and access to an official onion service. Further independent practice would strengthen these introductory skills.

References

[1] EC-Council. *Free Online Cyber Security Courses with Certificates*. Accessed 3 November 2026. https://www.eccouncil.org/cybersecurity-exchange/cybersecurity-exchange/cyber-novice/free-cybersecurity-courses-beginners/

[2] Tor Project. *About Tor Browser*. Accessed 3 November 2026. https://support.torproject.org/tor-browser/getting-started/about-tor-browser/

[3] Tor Project. *How do Onion Services work.* Accessed 3 November 2026. https://community.torproject.org/onion-services/overview/index.html

[4] Bitcoin.org. *Protect your privacy.* Accessed 3 November 2026. https://bitcoin.org/en/protect-your-privacy

[5] CodeRed. Validation of Course Completion issued to Mohamed Hussein for Introduction to Dark Web, Anonymity, and Cryptocurrency. Certificate 521166, issued 19 August 2026. Student-supplied image reproduced in Figure 1.

[6] Tor Project. Official onion service directory. Accessed 3 November 2026. https://onion.torproject.org/. Connection verification resource: https://check.torproject.org/. Figures 2 to 5 are original captures from the later practical demonstration.